

CASE STUDY / CISCO ACTIVE COMPONENTS • DESIGN • IMPLEMENTATION

A connected plant. An engineered network.

Netafim Irrigation India Pvt. Ltd.

Cisco switching and controller-based wireless, supported by a detailed Level 4 implementation workflow for configuration, testing and handover.

MANJUSAR / SAVLI • VADODARA

Client facility photograph supplied for this case study

13**ACCESS SWITCHES**

6 PoE + 7 data switches

1**CORE / AGGREGATION**

Catalyst 3750-X

11**WIRELESS APs**

Aironet 1602I

15**AP LICENCE PACKAGE**

1 Cisco 2504 controller

Quantities are ordered equipment in two POs dated 16 July 2014. Detailed topology and Level 4 activities are engineering reference views; as-built configurations and completion test records were not supplied.

The engagement | Active LAN and managed wireless

The active network connects wired users and business services through Cisco access switching, while a wireless controller manages the access points. D-TECH’s purchase-order scope covers supply and commissioning at Netafim’s Manjusr / Savli delivery location. This document develops that scope into a clear configuration and acceptance framework.

Component / model	PO 2900171536	PO 2900171540	Total / design role
WS-C2960S-48LPS-L · 48-port PoE, 370 W	6	-	6 · Powered endpoint access
WS-C2960S-48TS-L · 48-port data switch	6	1	7 · Wired data access
WS-C3750X-24T-L · 24-port data switch	1	-	1 · Core / aggregation reference
AIR-CT2504-15-K9 · Controller package	1	-	1 · Central WLAN management
AIR-CAP1602I-N-K9 · Controller-based AP	10	1	11 · Wireless access
802.3af injector · Separate line items	10	1	11 · Verify packed accessories

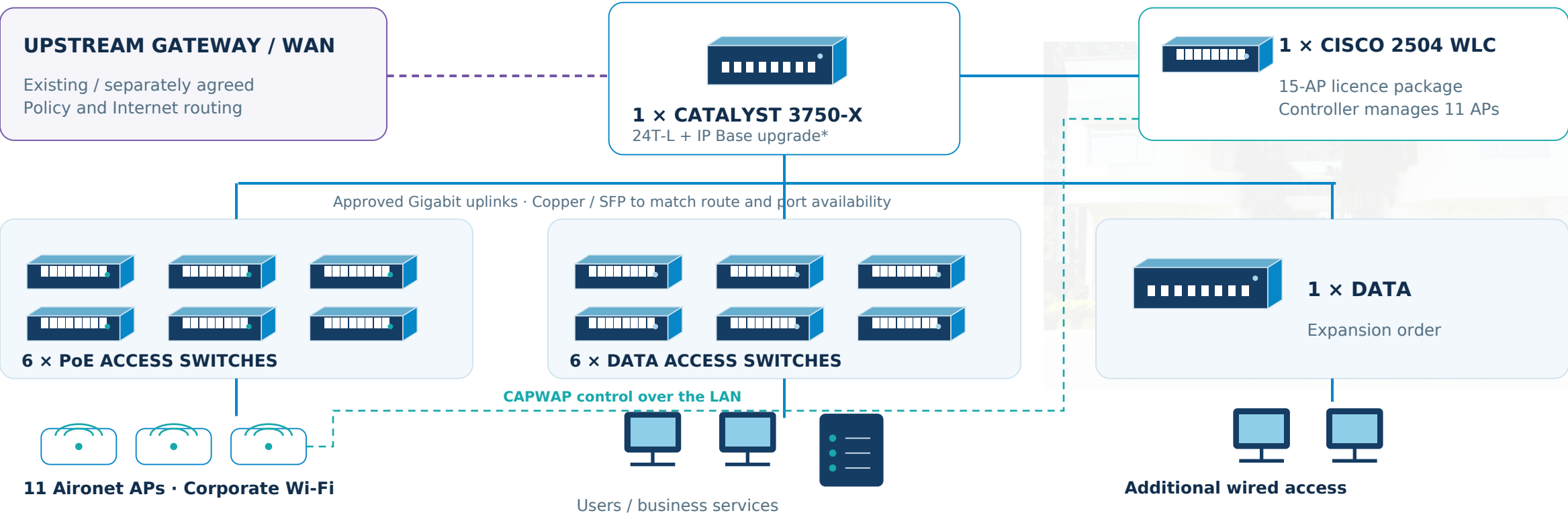
Included order details

SMARTnet service lines; 3750-X secondary PSU and 1G network module; rack accessories; an IP Base upgrade licence in the 3750-X configuration; installation and commissioning in PO 2900171536. AP configuration text also mentions an injector, so packed accessory quantities must be reconciled.

Inside: Logical topology 03 | Rack & uplinks 04 | VLAN / wireless design 05 | Level 4 workflow 06 | Acceptance tests 07 | Client / references 08

Cisco network design | Packet Tracer-style topology

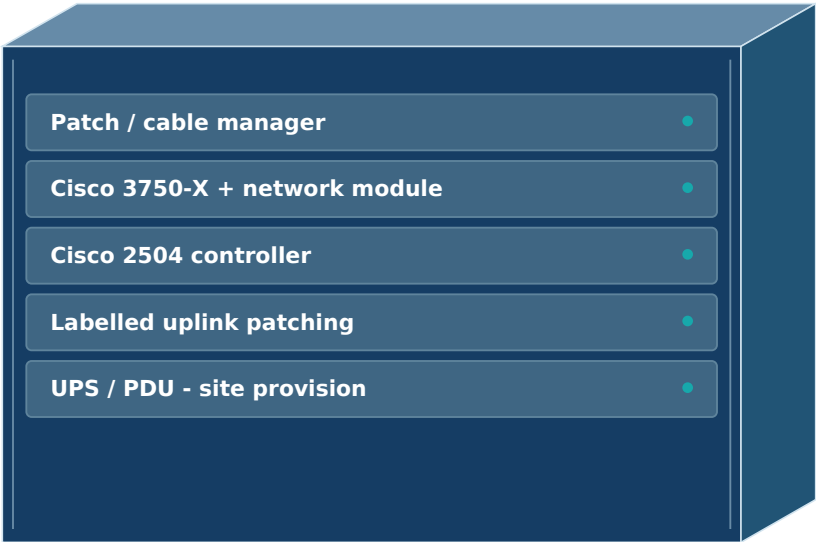
Reference allocation of the ordered devices. Switch locations, port assignments and uplink media follow the approved survey.



* Verify IP Base activation before using inter-VLAN routing. Illustrative endpoints and upstream equipment are outside these PO counts.

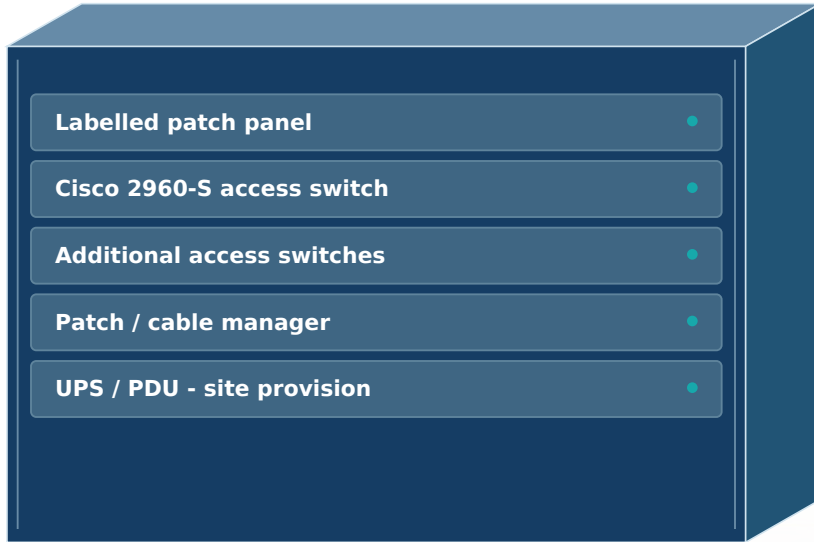
Physical deployment | Rack, power and uplink design

MDF / core reference



Rack positions are illustrative.
Final MDF / IDF allocation is survey-based.

IDF / access reference



Confirm rack space, airflow, earthing,
power segregation and service clearance.

GIGABIT UPLINK

Copper or compatible
1G optical link

Port and capacity planning

13 access switches provide 624 nominal RJ45 ports: 288 PoE-capable and 336 data ports. Actual usable endpoint ports depend on uplink allocation. The 3750-X 1G module provides four SFP slots; more than four direct optical access uplinks require a revised aggregation design or additional hardware.

Segmentation | Clear traffic groups and controlled access

Example VLAN	Purpose	Example subnet	Access policy
10 · Wired users	Business PCs / approved services	172.16.10.0/24	Reach approved applications; restrict management access.
20 · Corporate WLAN	Authenticated wireless users	172.16.20.0/24	SSID maps to approved VLAN and authentication policy.
40 · Guest WLAN	Guest connectivity if required	172.16.40.0/24	Internet-only policy at the approved gateway; deny internal access.
99 · Management	Switches, WLC and infrastructure	172.16.99.0/24	Authorised IT administration and approved monitoring sources.

Wireless packet path | Central switching reference



VLAN gateways may reside on the 3750-X after licence / feature validation or on an approved upstream gateway. AP management traffic and centrally switched WLAN client VLANs require the correct access-port and WLC-uplink configuration.

Addresses, SSIDs and access rules above are examples. The deployed IP plan, authentication service and software compatibility require client approval.

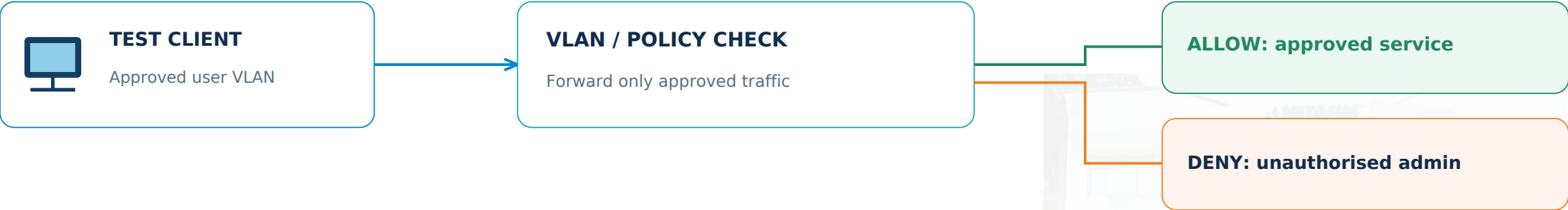
Level 4 installation | Engineering through handover

Level 4 describes the detailed engineering and troubleshooting scope used in this document. The work is completed against an approved configuration plan, with backups and a rollback method before the production cutover.

Step	Implementation activity	Deliverable / control
01 · Survey & baseline	Map switch locations, cable routes, uplink distances, endpoint demand and existing IP / VLAN plans.	Approved site and port schedule.
02 · Pre-stage	Record serials and accessories; validate IOS, WLC / AP compatibility, regulatory domain and licences.	Inventory, software baseline and licence checks.
03 · Install & label	Mount active devices; connect power and earth as specified; terminate and label approved uplinks.	Rack / port labels and link test record.
04 · Configure LAN	Apply VLANs, access ports, allowed trunk VLANs and the approved STP root. Add edge protections on suitable endpoint ports.	Reviewed switch configuration and backup.
05 · Commission Wi-Fi	Join APs to the 2504; map SSIDs to VLANs; configure approved authentication, channels and power.	AP join register, SSID and RF validation.
06 · Secure administration	Use named IT accounts, SSH, authorised management sources, time sync and supported monitoring / logs.	Access policy and management test.
07 · Cut over & validate	Test allowed and denied traffic, PoE draw, cable / uplink performance, user access and rollback readiness.	Acceptance checklist and exception log.
08 · Handover & support	Provide as-built diagram, port / IP register, configuration backups, operation notes and support contacts.	Signed handover against agreed scope.

Apply PortFast / BPDU Guard on approved edge ports. Preserve STP on switch uplinks. Verify stacking and redundancy hardware separately.

Commissioning | Prove the permitted packet path



Test area	Evidence to collect / example read-only checks
Layer 1 / uplinks	Check negotiated speed, errors and cable / optic condition; show interfaces status and interface counters.
VLAN & trunks	Check endpoint VLAN and permitted trunk VLANs; show vlan brief and show interfaces trunk.
Loop prevention & power	Verify the intended STP root and stable links; show spanning-tree. Check real PoE draw with show power inline.
Wireless service	Validate AP join, client authentication, DHCP, application access and walk-test coverage; WLC show ap summary / show wlan summary.
Isolation & recovery	Confirm guest-to-internal denial and authorised management access. Test the approved recovery / rollback scenario and compare the saved configuration.

Record actual results during commissioning. This is an authored Packet Tracer-style design and test plan; no simulation run or measured performance result is claimed. Validate command availability against the installed software.

Netafim | A structured delivery and support approach



Project delivery location

Netafim Irrigation India Pvt. Ltd.
Plot Nos. 268-270, 271B, GIDC Manjusar
Savli, Vadodara, Gujarat

Order scope and service record

POs 2900171536 and 2900171540, both dated 16 July 2014, establish the active-component supply and commissioning scope. SMARTnet entries are historical order items; current service entitlement must be checked separately.

Value of the engineering approach

A clear hierarchy of core, access and WLAN services; documented port allocation; controlled user and management traffic; and repeatable acceptance checks for operational handover.

Technical references | Cisco official documentation

Catalyst 2960-S specification: port / PoE / uplink capability
Catalyst 3750-X feature sets and network module capability
2960-S spanning-tree edge protections and configuration guidance

D-TECH | Design • Supply • Installation • Configuration • Commissioning • Support

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